1. Given Data

Object mass: **m = 200 g = 0.2 kg**

Gravity: **g = 9.81 m/s²**

From your CAD:

Radius ≈ **14.538 mm = 0.0145 m**

2. Calculate Object Weight (Force)

We use:

$$F = m \cdot g$$

So:

$$F = 0.2 \times 9.81 = 1.962 \text{ N}$$

This is the **minimum force needed to just hold the object (no safety)**

3. Apply Safety Factor

For grippers, we **always multiply** because of:

Slipping

Vibrations

Imperfect contact

Typical safety factor = **2 to 4**

Let's use **3**:

$$F_{required} = 1.962 \times 3 \approx 5.9 \text{ N}$$

4. Convert to Gripping Force per Finger

Since your gripper has **2 jaws**:

$$F_{per\ jaw} = \frac{5.9}{2} \approx 2.95 \text{ N}$$

5. Include Friction (VERY IMPORTANT)

Gripping depends on friction:

$$F_{grip} = \frac{F_{load}}{\mu}$$

assume:

Rubber contact → μ ≈ **0.5** (reasonable)

$$F_{grip} = \frac{5.9}{0.5} = 11.8 \text{ N}$$

total gripping force needed = ~12 N

6. Convert Force → Required Pressure
Now the key formula:

$$P = \frac{F}{A}$$

Where:
A = piston area

Assume Cylinder Size
Let's assume a **typical small pneumatic cylinder**:
Diameter = **20 mm → radius = 10 mm = 0.01 m**

$$A = \pi r^2 = \pi(0.01)^2 = 3.14 \times 10^{-4}\, m^2$$

Calculate Pressure

$$P = \frac{12}{3.14 \times 10^{-4}} \approx 38216\, Pa$$
$$P \approx 0.038\, MPa \approx 0.38\, bar$$